LAB ASSIGNMNET 14.3

Subject : AI Assistant Coding

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Batch : 50

Date : 18/3/26

Task-1 :

Prompt : Analyse the given scenario and generate a semantic HTML structure with sections: Home, About, Projects, Contact and Suggest responsive CSS using Flexbox or Grid and Customize CSS to add: Hover effects on project cards Smooth scrolling navigation.

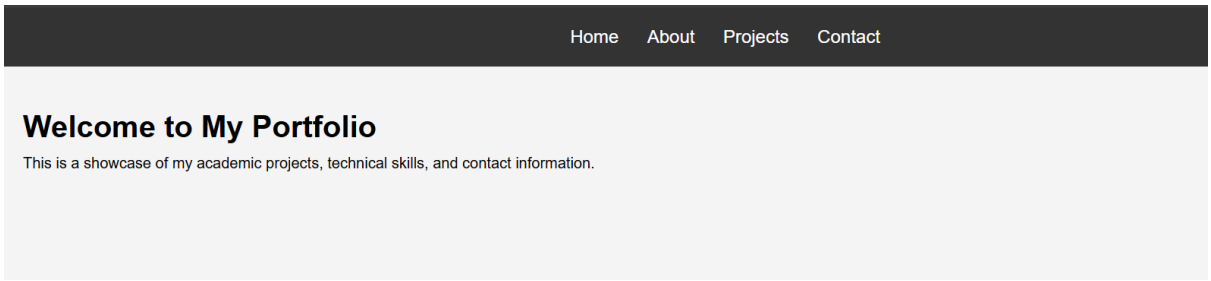
```html
<!-- A student wants to showcase academic projects, technical skills, and contact
information through a personal portfolio website. AI tools are used to speed up
development while maintaining quality and customization. -->
<!-- Analyze the given scenario and generate a semantic HTML structure with sections: Home, About, Projects, Contact -->
<!-- Suggest responsive CSS using Flexbox or Grid and Customize CSS to add: Hover effects on project cards Smooth scrolling navigation-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Portfolio</title>
    <style>
        /* Basic Reset */
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: Arial, sans-serif;
            line-height: 1.6;
        }

        /* Navigation */
        nav {
            background-color: #333;
            color: #fff;
            padding: 1rem;
            position: fixed;
            width: 100%;
            top: 0;
            z-index: 1000;
        }

        nav ul {
            display: flex;
            justify-content: center;
            list-style-type: none;
        }

        nav ul li {
            margin: 0 15px;
        }
```

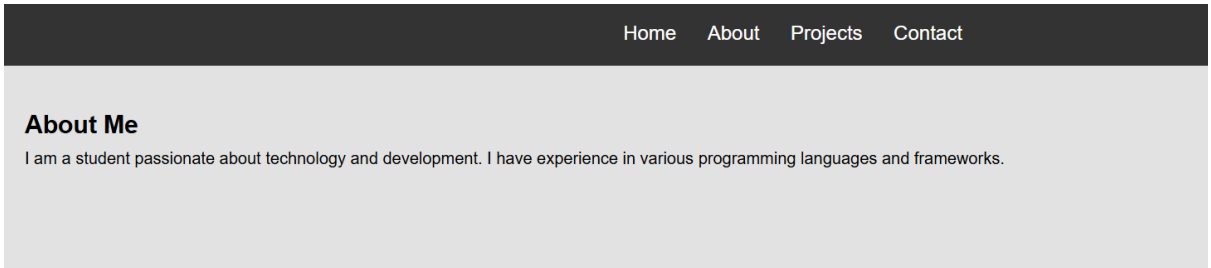
```html
<html lang="en">
<head>
    <style>
        nav ul li a {
            color: #fff;
            text-decoration: none;
            font-size: 1.2rem;
        }

        /* Sections */
        section {
            padding: 100px 20px; /* Add padding to account for fixed nav */
            min-height: 100vh;
        }

        #home {
            background-color: #f4f4f4;
        }

        #about {
            background-color: #e2e2e2;
        }

        #projects {
            background-color: #d0d0d0;
        }

        #contact {
            background-color: #c0c0c0;
        }

        /* Projects Grid */
        .projects-grid {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
            gap: 20px;
        }

        .project-card {
            background-color: #fff;
            padding: 20px;
            border-radius: 5px;
            box-shadow: 0 2px 5px rgba(0,0,0,0.1);
            transition: transform 0.3s ease, box-shadow 0.3s ease; /* Hover effect */
```

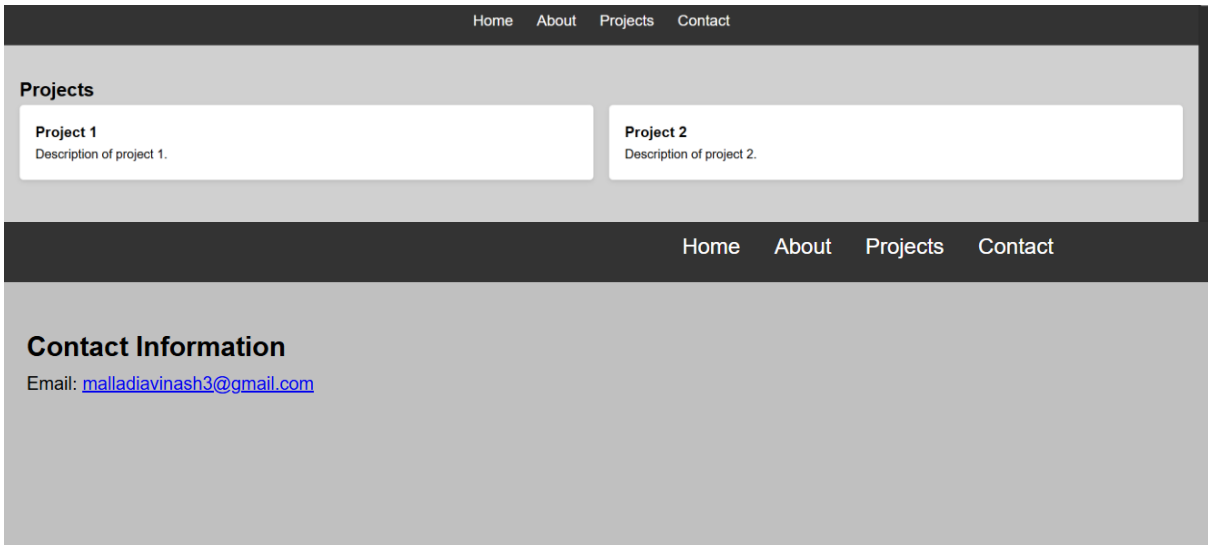
```html
<html lang="en">
<head>
    <style>
        .project-card:hover {
            transform: translateY(-10px);
            box-shadow: 0 4px 10px rgba(0,0,0,0.2);
        }
    </style>
</head>
<body>
    <nav>
        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#projects">Projects</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>

    <section id="home">
        <h1>Welcome to My Portfolio</h1>
        <p>This is a showcase of my academic projects, technical skills, and contact information.</p>
    </section>

    <section id="about">
        <h2>About Me</h2>
        <p>I am a student passionate about technology and development. I have experience in various programming languages and frameworks.</p>
    </section>

    <section id="projects">
        <h2>Projects</h2>
        <div class="projects-grid">
            <div class="project-card">
                <h3>Project 1</h3>
                <p>Description of project 1.</p>
            </div>
            <div class="project-card">
                <h3>Project 2</h3>
                <p>Description of project 2.</p>
            </div>
            <!-- Add more project cards as needed -->
        </div>
    </section>
```

```html
            <!-- Add more project cards as needed -->
        </div>
    </section>

    <section id="contact">
        <h2>Contact Information</h2>
        <p>Email:
            <a href="mailto: malladiavinash3@gmail.com">
                malladiavinash3@gmail.com
            </a>
        </p>
    </section>
</html>
```

Output :







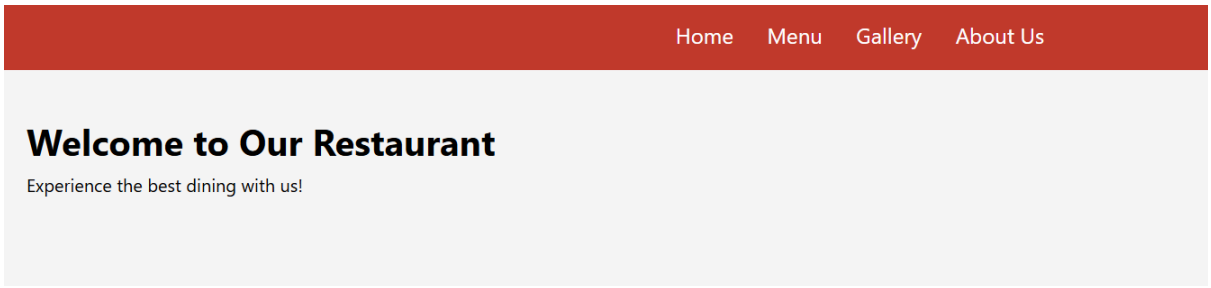
Task-2 :

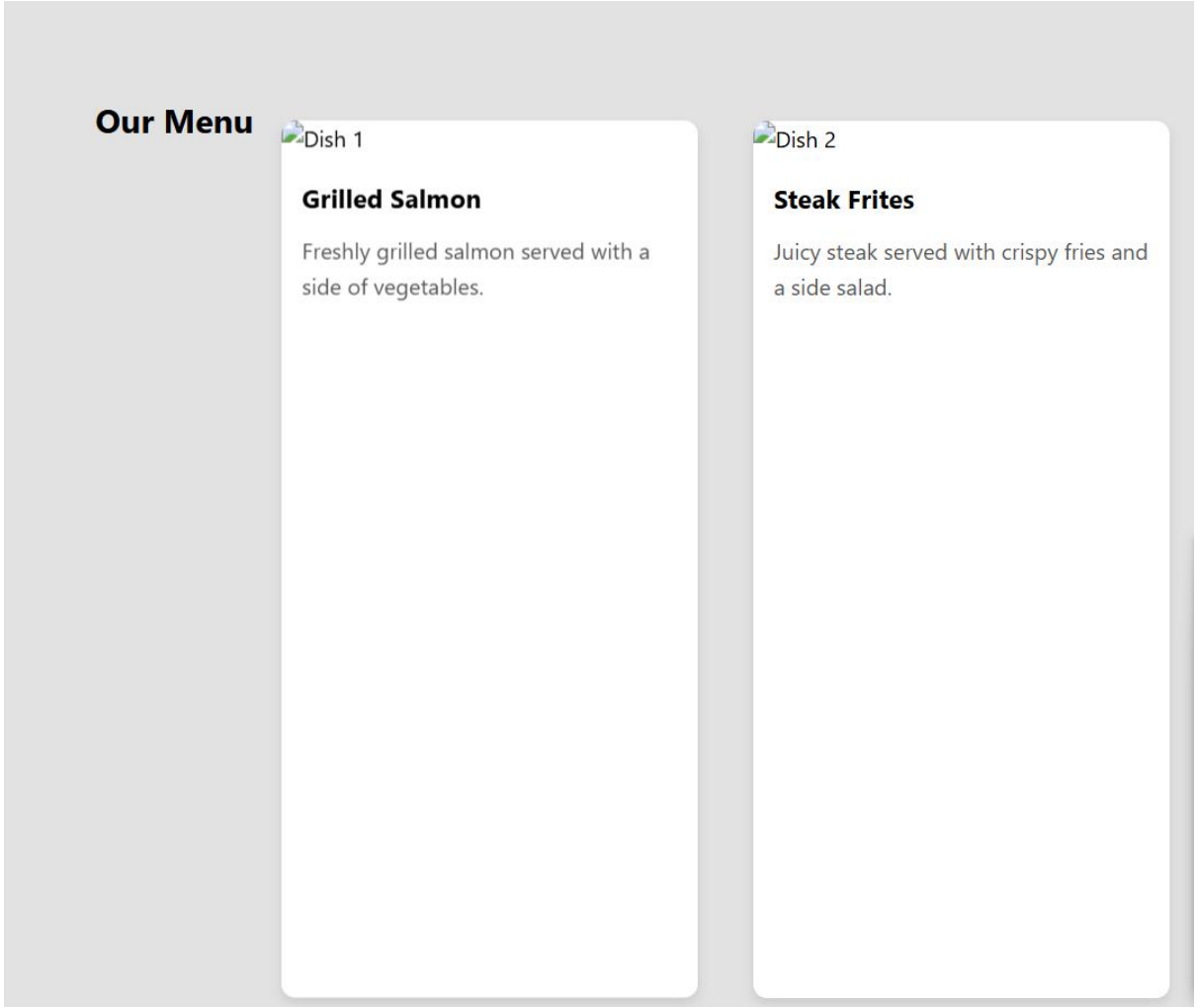
Prompt : Analyse the scenario and create a a navigation bar with links Home, Menu, Gallery, About Us. Generate a professional menu section styled with CSS cards.
Use AI Implement a JavaScript-based image slider for the gallery with real images from web/internet and Customize animations or transitions for better user experience.

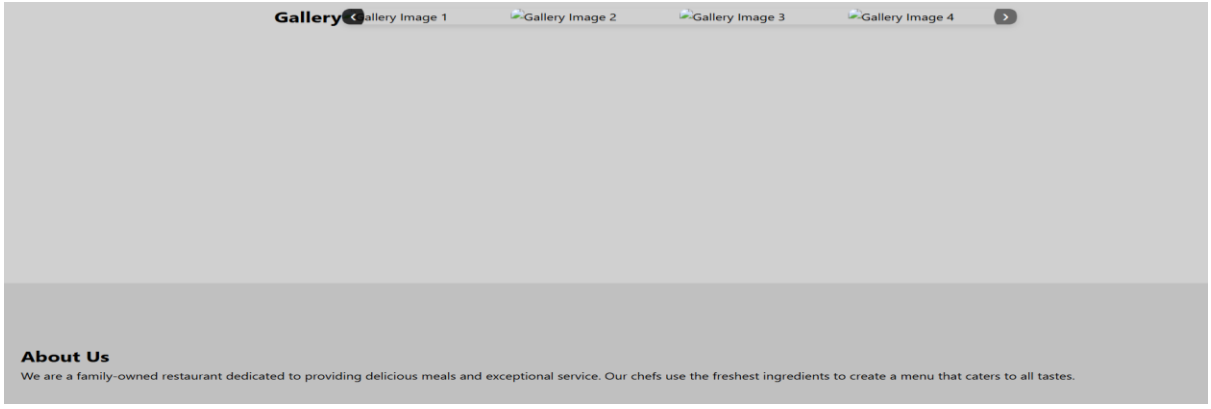
```html
<!-- A local restaurant requires a simple yet attractive landing page to display its
offerings. The developer uses AI assistance to rapidly generate UI components. -->
<!-- Analyze the scenario and create a a navigation bar with links Home, Menu, Gallery, About Us -->
<!-- Generate a professional menu section styled with CSS cards.
Use AI Implement a JavaScript-based image slider for the gallery with real images from web/internet.-->
<!-- Customize animations or transitions for better user experience.-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Landing Page</title>
    <style>
        /* Basic Reset */
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            line-height: 1.6;
        }

        /* Navigation */
        nav {
            background-color: #c0392b;
            color: #fff;
            padding: 1rem;
            position: fixed;
            width: 100%;
            top: 0;
            z-index: 1000;
        }

        nav ul {
            display: flex;
            justify-content: center;
            list-style-type: none;
        }

        nav ul li {
            margin: 0 15px;
```

```html
    <section id="home">
        <h1>Welcome to Our Restaurant</h1>
        <p>Experience the best dining with us!</p>
    </section>

    <section id="menu">
        <h2>Our Menu</h2>
        <div class="card">
            <img src="https://via.placeholder.com/300x200" alt="Dish 1">
            <div class="card-body">
                <h3>Grilled Salmon</h3>
                <p>Freshly grilled salmon served with a side of vegetables.</p>
            </div>
        </div>
        <div class="card">
            <img src="https://via.placeholder.com/300x200" alt="Dish 2">
            <div class="card-body">
                <h3>Steak Frites</h3>
                <p>Juicy steak served with crispy fries and a side salad.</p>
            </div>
        </div>
        <!-- Add more cards as needed -->
    </section>

    <section id="gallery">
        <h2>Gallery</h2>
        <div class="slider">
            <div class="slides">
                <img src="https://via.placeholder.com/800x400" alt="Gallery Image 1">
                <img src="https://via.placeholder.com/800x400" alt="Gallery Image 2">
                <!-- Real images from Instagram -->
                <img src="https://via.placeholder.com/800x400" alt="Gallery Image 3">
                <img src="https://via.placeholder.com/800x400" alt="Gallery Image 4">
            </div>
            <div class="slider-nav
">
                <button id="prev">&#10094;</button>
                <button id="next">&#10095;</button>
            </div>
        </div>
    </section>
```

Output :


Home
Menu
Gallery
About Us
Welcome to Our Restaurant
Experience the best dining with us!


Our Menu
Dish 1
Grilled Salmon
Freshly grilled salmon served with a side of vegetables.
Dish 2
Steak Frites
Juicy steak served with crispy fries and a side salad.


Gallery
Gallery Image 1
Gallery Image 2
Gallery Image 3
Gallery Image 4
About Us
We are a family-owned restaurant dedicated to providing delicious meals and exceptional service. Our chefs use the freshest ingredients to create a menu that caters to all tastes.

Task-3:

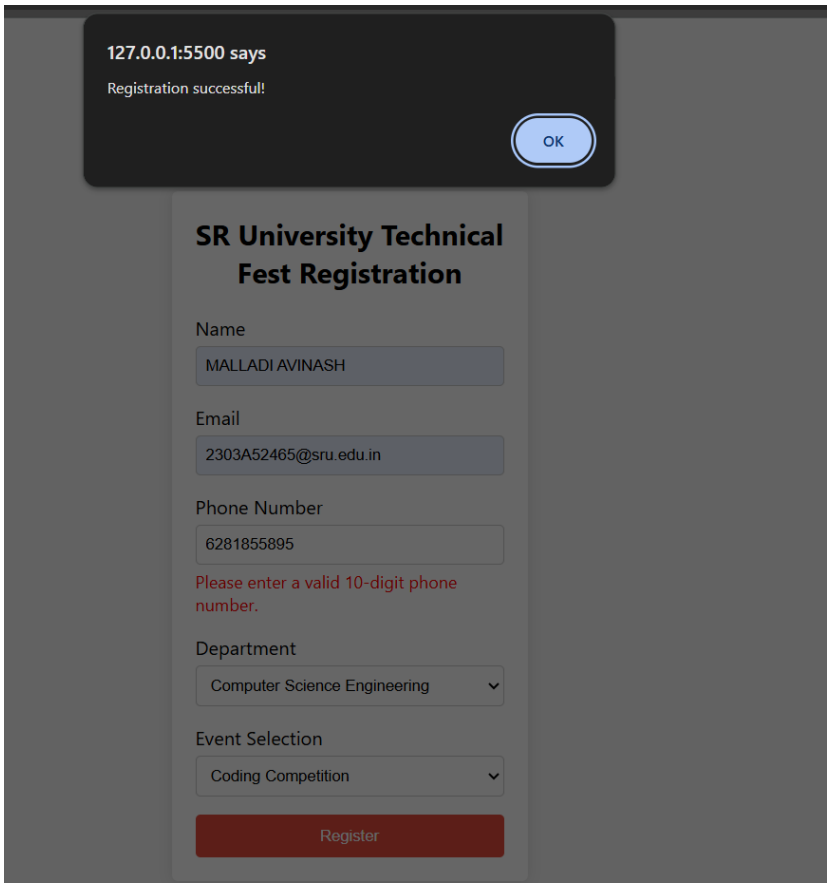
Prompt : Analyse the above given scenario and generate an HTML form with the following fields: Name, Email, Phone Number, Department, Event Selection. Style the form using CSS for readability and accessibility. Use AI to Implement JavaScript validation : Email format validation, Phone number length validation, Required field checks in this file itself,
I need User-friendly and visually appealing form, Real-time validation messages, Prevention of invalid submissions.

```html
<!-- SR University is hosting a technical fest and requires an online registration form
with real-time validation to ensure accurate user input -->
<!-- Analyse the above given scenario and generate an HTML form with the following fields: Name, Email, Phone Number, Department, Event Selection-
<!-- Style the form using CSS for readability and accessibility.-->
<!-- Use AI to Implement JavaScript validation : Email format validation, Phone number length validation, Required field checks in this file itsel
<!-- I need User-friendly and visually appealing form, Real-time validation messages, Prevention of invalid submissions-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>SR University Technical Fest Registration</title>
    <style>
        /* Basic Reset */
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background-color: #f4f4f4;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .registration-form {
            background-color: #fff;
            padding: 20px;
            border-radius: 5px;
            box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
            width: 300px;
        }

        .registration-form h2 {
            text-align: center;
            margin-bottom: 20px;
        }
```

```html
<body>
    <form class="registration-form" id="registrationForm">
        <h2>SR University Technical Fest Registration</h2>
        <div class="form-group">
            <label for="name">Name</label>
            <input type="text" id="name" name="name" required>
            <div class="error-message" id="nameError"></div>
        </div>
        <div class="form-group">
            <label for="email">Email</label>
            <input type="email" id="email" name="email" required>
            <div class="error-message" id="emailError"></div>
        </div>
        <div class="form-group">
            <label for="phone">Phone Number</label>
            <input type="text" id="phone" name="phone" required>
            <div class="error-message" id="phoneError"></div>
        </div>
        <div class="form-group">
            <label for="department">Department</label>
            <select id="department" name="department" required>
                <option value="">Select Department</option>
                <option value="CSE">Computer Science Engineering</option>
                <option value="ECE">Electronics and Communication Engineering</option>
                <option value="ME">Mechanical Engineering</option>
                <!-- Add more departments as needed -->
            </select>
            <div class="error-message" id="departmentError"></div>
        </div>
        <div class="form-group">
            <label for="event">Event Selection</label>
            <select id="event" name="event" required>
                <option value="">Select Event</option>
                <option value="coding">Coding Competition</option>
                <option value="robotics">Robotics Challenge</option>
                <option value="quiz">Tech Quiz</option>
                <!-- Add more events as needed -->
            </select>
```

```
form.addEventListener('submit', function(e) {
    e.preventDefault();
    let isValid = true;

    // Name validation
    if (nameInput.value.trim() === '') {
        nameError.textContent = 'Name is required.';
        isValid = false;
    } else {
        nameError.textContent = '';
    }

    // Email validation
    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailPattern.test(emailInput.value.trim())) {
        emailError.textContent = 'Please enter a valid email address.';
        isValid = false;
    } else {
        emailError.textContent = '';
    }

    // Phone number validation
    if (phoneInput.value.trim().length !== 10 || isNaN(phoneInput.value.trim())) {
        phoneError.textContent = 'Please enter a valid 10-digit phone number.';
        isValid = false;
    } else {
        phoneError.textContent = '';
    }

    // Department validation
    if (departmentSelect.value === '') {
        departmentError.textContent = 'Please select a department.';
        isValid = false;
    } else {
        departmentError.textContent = '';
    }
```

Output :



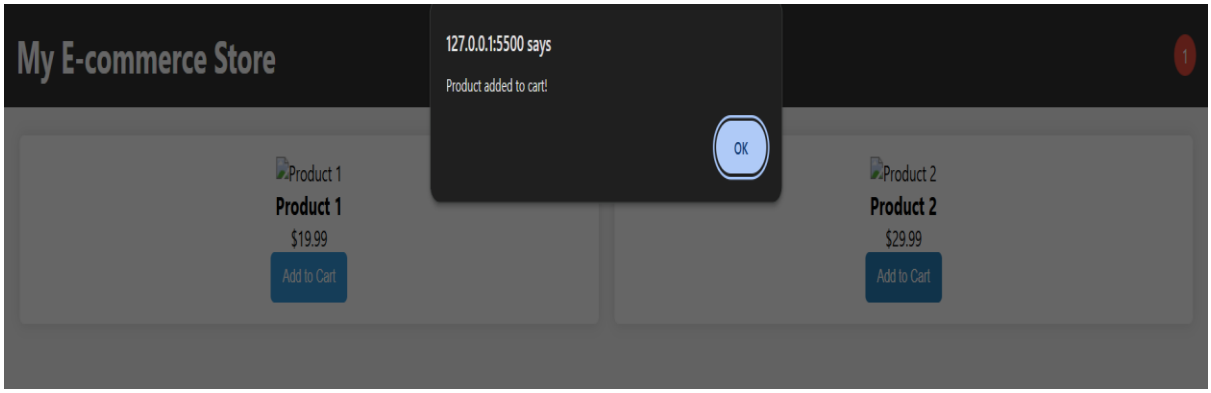
Task-4 :

Prompt : Analyse the above scenario and generate a grid-based product catalog using HTML and CSS in this file itself. Implement an “Add to Cart” feature using JavaScript.
Modify generated code to include: Cart item counter at the top-right corner, Visual feedback when a product is added. I need Functional product catalog, Working cart counter, Interactive user experience for this task with full classic and professional style like a working online store.

```html
<!-- A startup is developing a basic e-commerce product page to showcase products
and manage simple cart functionality. -->
<!-- Analyze the above scenario and generate a grid-based product catalog using HTML and CSS in this file itself. -->
<!-- Implement an “Add to Cart” feature using JavaScript.-->
<!-- Modify generated code to include: Cart item counter at the top-right corner, Visual feedback when a product is added-->
<!-- I need Functional product catalog, Working cart counter, Interactive user experience for this task with full classic and professional
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>E-commerce Product Page</title>
    <style>
        /* Basic Reset */
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background-color: #f4f4f4;
        }

        /* Header */
        header {
            background-color: #333;
            color: #fff;
            padding: 1rem;
            display: flex;
            justify-content: space-between;
            align-items: center;
        }

        .cart-counter {
            background-color: #e74c3c;
            color: #fff;
            border-radius: 50%;
            padding: 5px 10px;
```

```style
<style>
    .cart-counter {
        color: #fff;
        border-radius: 50%;
        padding: 5px 10px;
        font-size: 0.9rem;
    }

    /* Product Catalog */
    .product-catalog {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
        gap: 20px;
        padding: 20px;
    }
```

```
        .product-card {
            background-color: #fff;
            padding: 15px;
            border-radius: 5px;
            box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
            text-align: center;
        }

        .product-card img {
            max-width: 100%;
            height: auto;
        }

        .add-to-cart-btn {
            background-color: #3498db;
            color: #fff;
            border: none;
            padding: 10px 15px;
            border-radius: 5px;
            cursor: pointer;
        }

        .add-to-cart-btn:hover {
            background-color: #2980b9;
        }
    </style>
</head>
<body>
    <header>
        <h1>My E-commerce Store</h1>
        <div class="cart-counter" id="cartCounter">0</div>
    </header>

    <div class="product-catalog">
        <div class="product-card">
            <img src="https://via.placeholder.com/150" alt="Product 1">
            <h3>Product 1</h3>
```

```
            <img src="https://via.placeholder.com/150" alt="Product 2">
            <h3>Product 2</h3>
            <p>$29.99</p>
            <button class="add-to-cart-btn" onclick="addToCart()">Add to Cart</button>
        </div>
        <!-- Add more products as needed -->
    </div>

    <script>
        let cartCount = 0;

        function addToCart() {
            cartCount++;
            document.getElementById('cartCounter').textContent = cartCount;
            alert('Product added to cart!');
        }
    </script>
</body>
</html>
```

Output :